

Research Paper

Kriging Empirical Mode Decomposition via support vector machine learning technique for autonomous operation diagnosing of CHP in microgrid



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HIGHLIGHTS

- Negligible none detection zone for CHP generator.
- Avoiding of threshold selection.
- Suitable for microgrid application.
- Usable in noisy condition.
- Best data selection method.

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ABSTRACT

Combined Heat and Power (CHP) is the one of new energy resources which has been added to power system in recent years. High efficiency, Loss reducing of power system and etc, are the main advantages of CHP as same as other distributed generations. But, unwanted islanding is one of the main problems for this generation. This article presents a novel technique for CHP unit islanding detection using Kriging Empirical Mode Decomposition (KEMD) and Support Vector Machine (SVM) pattern learning technique. In this technique the variation of Intrinsic Mode Functions (IMF) of local signals in two-dimensional mode is utilized as input data of relay. An optimal signal selection model is applied to the proposed relay in order to Non-Detection Zone (NDZ) and fails detection reducing. The best signal selection is introduces based on mean square value between islanding and non-islanding conditions. Also, by considering Optimal SVM model for the proposed relay as a pattern recognizing and weighing it using shark smell optimization, this technique has overcome the threshold selection problem. This relay is applied to CHP system in a microgrid system contains various types of DGs. Many islanding and non-islanding situation in various operation conditions in the studied microgrid are simulated. The results of simulation results are show that the proposed relay is suitable for microgrid application. Negligible NDZ, high detection time, zero fail detection and low cost of this relay are the main advantages of the proposed technique.

1. Introduction

With the increasing energy consumption, environmental carbon release and depletion of fossil fuels, Energy manufacturers tend to increase the use of renewable energy such as wind power, solar

photovoltaic and landfill gas [1]. Individual Distributed Generation (DG) connected to the power system has intermittence, randomness and the uncertainty character, which are difficult to troubleshoot in some cases in power systems [2]. In order to solve these problems, some solutions are suggested in microgrid [3]. The microgrid is a subsystem

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Nomenclature		P	Active power
Indices		Q	Reactive power
		I	Current
		L_m	DG inductance
		Variables	
		ΔP	Active power variation
		ΔQ	Reactive power variation
		$\Delta \delta$	Phase angle variation
		ΔV_d	d-axes voltage variation
		ΔV_q	q-axes voltage variation
		THD_V	Total harmonic distortion of voltage
		THD_I	Total harmonic distortion of current
		H_1	Low pass filter
		H_2	High pass filter
		C_0	Scale factor
		d_j	j-th level wavelet factor
		E_{c0}	Scale factor energy
		E_{d_j}	j-th level wavelet factor energy
		ISE_{dk}	average square error of islanding signals
Parameters			
		f_n	Nominal frequency
		V_t	Terminal Voltage
		Δf	Frequency variation
		ΔV_t	Terminal voltage variation

of power system that consists of multiple DG units, storage devices and loads which that can be activated in both grid-connected and islanded (autonomous) modes [4]. If all or part of the microgrid, including load and DG, is separated from the main grid and DGs remain, the island state has occurred [5].

Islanding conditions are being planned or undesirable [6]. In the planned islanding conditions, it is possible to provide active/reactive power for MG loads using by DG units and there is no active/reactive power interchange with the grid. As a result, island conditions operate without any transients, and the MG continues performing like an independent system [7]. In some cases, island conditions occur because of disturbances in the system. If the unwanted island conditions do not occur, the operating conditions of the MG depend on the delivery of electrical loads between DG units or importing or exporting energy from/to the grid can be altered. These conditions must be identified in order to control the sources of MG, when island conditions occur.

In general, in order to identify unwanted island conditions, there are three possible solutions: passive, active and remote techniques [8]. The process of Remote operation [9] is based on the relationship between the utility and the DGs. The disadvantages of this method are high costs and the difficulty of implementing it. An active method cause disturbances in distributed systems and pushes the system in an island state towards unstable [10]. Because of the disadvantages of these methods and their performance in both grid-connected and autonomous conditions, both techniques are inappropriate for microgrid resources. Therefore, the mentioned methods are not useful for distribution systems [11].

The active method is proportional with the parameters of the power system (voltage, current, frequency and etc) and identifies the island mode using data processing. The active method is proportional with the parameters of the power system and identifies the island mode using data processing. A number of active techniques are: over/under voltage (OV/UV) and frequency (OF/UF) [12], total harmonic distortions [13], rate of change of frequency and rate of change of generator power output [14], phase displacement monitoring, vector surge [15], rate of change of phase angle derivation [16], absolute value of certain wavelet coefficient [17], Time-frequency detection algorithm [18], transient current and voltage signals through discrete wavelet transform [19,20]. It should be noted that the unwanted island state occurring in power balance operation, some passive islanding discovery approaches may not respond properly, or it takes a long time to identify the islands.

Also, one of the most important issues in the passive technique is the choice of threshold. Too low values of the threshold cause the nuisance tripping problem, and on the contrary, the high value of the threshold islanding may not be identify.

Recently, islanding diagnosing technique using wavelets models are widely applied to distributed generations for develop the performance of passive methods. Stability, flexibility, Simplicity and low cost to implementation are the main advantages of WT methods. some of WT methods are; WGM1.0 & Machine learning [21], WSE & matrix analysis [22], Haar & Neuro-Fuzzy [23], Daubechies db4 [24], WPT & dq frame analysis [25], Wavelet singular entropy [26]. In the mentioned methods, the signal selection method hasn't applied to detection process. Also, in the above methods the value of signal is considered which can be caused to mistake detection. Moreover, another main problem in passive algorithms is the choosing of the threshold. If this threshold be very low the wrong tripping makes challenge, otherwise with very high values of threshold, detection of islanding mode may not be done.

In this paper, the various local signals are measured in the Power Common Coupling (PCC) point of DG and these signals are sorted by a mathematical theory and the best of them is selected as input signal. At First, all of local signals are recorded under various islanding and non-islanding situation under every loading conditions of network. In next step, using Kriging Empirical Mode Decomposition (KEMD) in various decomposition levels, measured signals' are decomposed. Two criterions named islanding signals' standard deviation (ISSD) and cross islanding and non-islanding signal's standard deviation (CINSSD) are defined for optimal signal selection and all of signals are sorted using these criterions. The SVM based technique is designed for state variation of this signal and the relay is trained optimally using optimization algorithms and these signals are employed as input data. Totally, the proposed method has prominent features as bellow:

- Optimal signal selecting using two defined ISSD and CINSSD criterions.
- Choosing optimum window time and number of signals as input for relay.
- Negligible non-detection zone achievement and significant reduction noise sensitivity.
- Avoidance from defining threshold value by considering SVM.

The proposed technique is applied to CHP unit in study microgrid

system and the effectiveness and accuracy of the method is approved by the simulation results. Study system modeling is done in Section 2. Applied events to study system in defined in Section 3. Section 4 contains the proposed detection methodology. The numerical results is presented in Section 5 and in the final section, the conclusion is presented.

2. System description

The studied system in this paper is a microgrid contains various distributed generation with a state space mathematical model and NDZ opinion and formulation. With the aim of describing the function of the target algorithm in the NDZ regions, the size of decomposition processing window and the method of island identification are formulated.

2.1. Microgrid case study

The single-line diagram of the microgrid system to explore the proposed islanding detection algorithm is depicted in Fig. 1. The structure of the studied system consists of a distribution system connected to the utility grid via a 69 kV line and contains four DG units. DG1 and DG2 are voltage source converter base system with 300 kW rated power which are controlled by the active/reactive control strategy system. DG3 represents a wind turbine with self-excited asynchronous generator. DG4 represents a CHP generator with 500 kW rated power and controlled by excitation and governor system. The radial feeders supply linear and nonlinear loads of the system. The lines and linear loads like a RL branches Connected to the circuit in series in each phase is modeled. Non-linear loads are simulated as a three phase full controlled rectifier load and induction motors. B1 and B16 are composed of three phase capacitor banks; B16 is exploited as

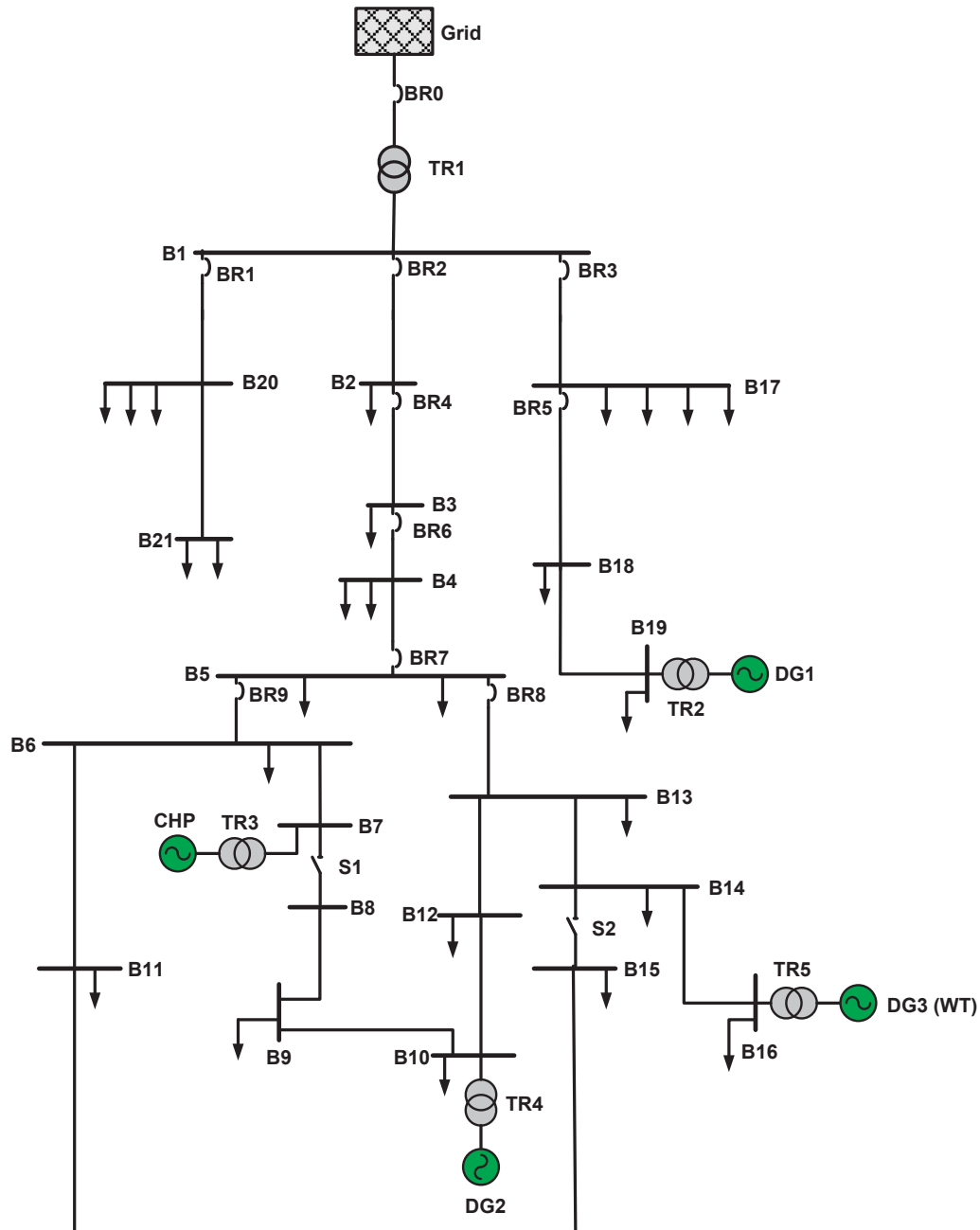


Fig. 1. Single line diagram of studied microgrid system.

asynchronous generator excitation system.

The system response for each 0.1 pu load changing in island state is depicted in Fig. 2. The study time for the system output in the suggested method is expressed using time transient. In this figure, the analysis time for all conditions is considered for all conditions with steps of 0.2 s (see Fig. 3).

2.2. Non-detection zone

One of the most important indicators for the recognition of the islanding identification method is non-detection zone. The time when the island's conditions are not identifiable in a timely manner is defined as the NDZ operational area. In this area, the difference between power generation and consumption is low. In this area, the difference in the amount of power produced and consumed is a small amount. The non-detection zone (NDZ) for active power is formulated as follow:

$$\Delta P = 3 \times V \times I - 3 \times (V + \Delta V) \times I = -3 \times \Delta V \times I \quad (1)$$

In the above equation, the nominal voltage and nominal current are shown respectively with V and I . Given that the acceptable voltage range is from 0.88 to 1.1, the active imbalance values are 71.5 kW and -65 kW, respectively. The NDZ for reactive power is formulated as follows:

$$\Delta Q = 3 \frac{V_t^2}{\omega_n L_m} \left(1 - \frac{f_n^2}{(f_n - \Delta f)^2} \right) \quad (2)$$

Also, since the frequency range is between 49.7 and 50.3 Hz, the amounts of reactive power imbalances are equal to 41.5 and 44, respectively.

2.3. Prescribed events

The suggested approach for identifying the island is obtained by analyzing the frequency time-frequency and learning in a stable variable. Some of the events are distinct according to the operational requirements in IEEE1547 standards and possible operating network to choose the appropriate variables and learn the proposed relay. In order to maintain the main characteristics of the system performance, events are simulated off-line, which is presented in the proposed method. In general, events are classified into two categories include: namely islanding and non-islanding, which are taken for the proposed islanding detection relays.

In small imbalance active and reactive power situations (NDZ), BR2, BR4, and BR7 are unlocked.

(a) In large imbalance active and reactive power, BR2 and BR7 are

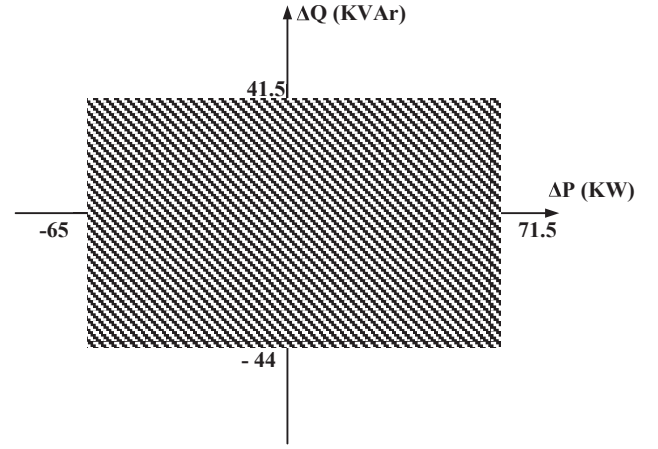


Fig. 3. NDZ for the study test system.

unlocked.

- (b) Tripping breakers BR7, BR8, and BR9 subsequent to a high resistance fault at bus B5,
- (c) Tripping breakers 'BR2' and 'BR4' subsequent to a high resistance fault at bus B2.

Events for non-islanding conditions are expressed as follows.

- (a) Motor operating with different capacities at buses B5 and B1
- (b) at buses B5 and B1, Capacitor bank switching with different capacities,
- (c) Non-linear load connecting to network at buses B5 and B1,
- (d) Tripping BR3 and disconnecting DG1,

In accordance with the number of functional loads and different settings of the testing system, consist of radial/ring structure of microgrid, all of the cases mentioned above are simulated. A total of 744 events have been simulated, of which 524 are island condition and 350 cases are related to match power conditions (NDZ), and 174 cases are associated to mismatch power conditions. Similarly, the non-island mode is also simulated on 220 cases.

3. Empirical Mode Decomposition (EMD)

The EMD purpose is parsing an unstable signal to smoother constituents named Intrinsic Mode Function (IMF). This intrinsic mode function is a vacillating function as matching part of harmonic function which has two key features as following: (1) the regional max, min and

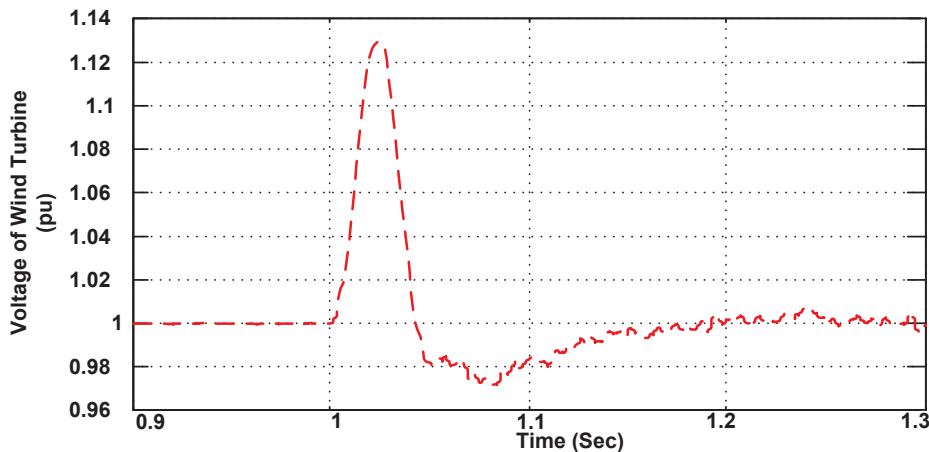


Fig. 2. Step response of system for microgrid DGs.

zero crossing numbers in this function is same, which means, their difference upper bound is one. (2) Average of this function is equal to zero.

EMD utilizes a repetitive sifting procedure for parsing signal to IM function elements. Steps of this process can be briefly described as follow:

EMD algorithm:

- (1) Compute the local maximum and minimum of the signal, indicated by $x(t)$, within the assumed domain.
- (2) Connect the maximum of $x(t)$ with an interpolating function, such as to an upper envelope $u(t)$ is generated around the signal. EMD utilizes spline of third-order, known as Cubic Spline (CS), for the interpolation [27]. Likely, connect the minimum of $x(t)$ with the interpolating function to generated a lower envelope $l(t)$ about the signal. Fig. 4 illustrated these envelopes.
- (3) The upper and lower envelopes average, i.e., $s(t) = [u(t) + l(t)]/2$, is computed, which is the curve of local average in Fig. 1.
- (4) If the stopping condition a is contented, the local averages $s(t)$ is an IMF and go to step 5; otherwise, replace the signal with signal minus the local average, i.e., $x(t) - s(t) \rightarrow x(t)$, and go back to step 2.
- (5) Assume n IMF components $s_1(t), s_2(t), \dots, s_n(t)$ have been created so far. Calculate the residual $r(t) = x(t) - [s_1(t) + s_2(t) + \dots + s_n(t)]$. If the stopping condition b is contented, terminate the algorithm; otherwise, replace the signal with the residual, i.e., $r(t) \rightarrow x(t)$, and go back to step 2.

Stop term states, the comparative variance among 2 consecutive local averages must be lower than a predetermined amount [28]. In other words, generated alteration by repetition of inner loop is ignorable in the local average. Stop term b for outer loop expresses, the residue must be uniformly increasing/reducing which means, in residual should not have variation in regarded domain [29]. Once, the mechanism is finished with n IMF elements, the signal is parsed as follow:

$$x(t) = \sum_{i=1}^n s_i(t) + r(t) \quad (3)$$

The EMD is a high performance instrument in signal processing for analyzing regional treatments. It analyzes the unstable signal via parsing that to element with lower volatility. Nevertheless, there is a significant challenge in this method that is production of spurious min and max points which are presented by CS profile fitting. The spurious extremes cause to spurious over and undershoot that lead to authenticity reduction in generated IMFs and rise in repetition of sifting procedure. Fig. 5 shows these oscillations. A developed EMD in suggested to tackle these disadvantages and improve the EMD efficiency. This modified EMD substituted the CS interpolation with IKIM. In next, subsequence to introduction of normal KIM, improved KIM (IKIM) is presented.

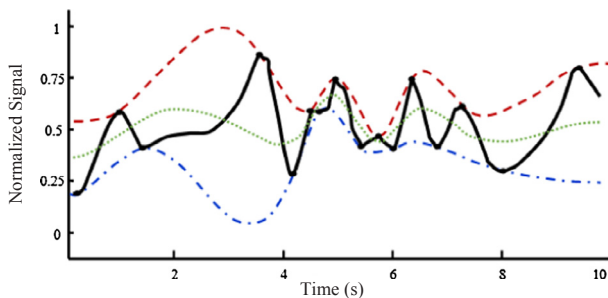


Fig. 4. Illustration of the sifting processes including original signal $x(t)$ (solid black line), upper envelope $u(t)$ (dashed red line), lower envelope $l(t)$ (dash-dot blue line) and local mean or Intrinsic Mode Function (IMF) $s(t)$ (dotted green line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

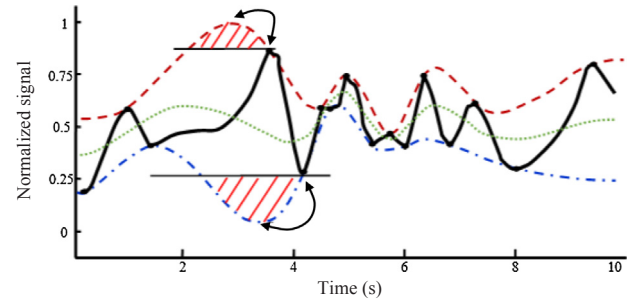


Fig. 5. Illustration of the Empirical Mode Decomposition (EMD) problem: Original signal (solid black line), upper envelope (dashed red line), lower envelope (dash-dot blue line), IMF (dotted green line) and wrong undershoots/overshoots (red hachure). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Then, the suggested IKIM is exerted to EMD.

Kriging is an interpolation approach that approximates amount of a function in a determined point as a weighted function amounts sum at adjacent points [30]. This method has significant merits in comparison to other interpolation methods. Kriging method, instead of assigning random function, allocates weights based on data driven method which leads to upper weights to insulated points contrasting to points in a cluster. Furthermore, this method in addition to parameter estimation can estimate also its error via kriging variance [31].

KIM via next relation, approximates amount of $x(t)$ time series at t point:

$$\hat{x}(t) = \sum_{i=1}^M w_i x(t_i), \quad \sum_{i=1}^M w_i = 1 \quad (4)$$

In this equation, $\hat{x}(t)$ is approximated value of $x(t)$ by KIM, and the adjacent points of t are determined by $t_i, 1 \leq i \leq M$.

The weights of w_i in normal KIM [32], are achieved via variance minimization method. Error of KIM is calculated as follow for $x(t)$ estimation:

$$e[x(t)] = \hat{x}(t) - x(t) = \sum_{i=1}^M w_i x(t_i) - x(t) \quad (5)$$

Also, it can be rewritten in following vector shape:

$$e[x(t)] = [w_1 \dots w_M - 1] [x(t_1) \dots x(t_M) x(t)]^T \quad (6)$$

In which, T denotes the transpose. Also, $\text{Var}(e[x(t)])$ is error variance that can be calculated by bilinear form theory as follows [33]:

$$\text{Var}(e[x(t)]) = [w_1 \dots w_M - 1] \cdot \text{Var}([x(t_1) \dots x(t_M) x(t)]^T) \cdot [w_1 \dots w_M - 1]^T \quad (7)$$

Eq. (5) can be rewritten, through parsing the vector $e[x(t)]$ as follows:

$$\text{Var}(e[x(t)]) = [w_1 \dots w_M - 1] \cdot \begin{bmatrix} \text{Var}(X) & \text{Cov}(X, x(t)) \\ [\text{Cov}(X, x(t))]^T & \text{Var}(x(t)) \end{bmatrix} \cdot [w_1 \dots w_M - 1]^T \quad (8)$$

In this formula, the covariance function is determined by $\text{Cov}(\dots)$. Subsequently, KIM indicates the w_i weights by resolving of below optimization equations:

$$\text{Min}_{w_1, \dots, w_M} \{ \text{Var}(e[x(t)]) \} \text{ st. } \sum_{i=1}^M w_i = 1 \quad (9)$$

In which, $\text{Var}(e[x(t)])$ introduced in Eq. (8). The ordinary KIM considers Gaussian trend for M sample (earlier distribution) in order to calculate variance/covariance of Eq. (8). Therefore, the outcomes of prior distribution become Gaussian [31]. Solution of represented optimization in Eq. (9) can be done by Lagrange multipliers' approach that results following answer [32]:

$$\begin{bmatrix} w_1 \\ \vdots \\ w_M \\ \lambda \end{bmatrix} = \begin{bmatrix} \text{Cov}(x(t_1), x(t_1)) & \cdots & \text{Cov}(x(t_1), x(t_M)) & 1 \\ \vdots & & & \\ \text{Cov}(x(t_M), x(t_1)) & \cdots & \text{Cov}(x(t_M), x(t_M)) & 1 \\ 1 & \cdots & 1 & 0 \end{bmatrix}^{-1} \cdot \begin{bmatrix} \text{Cov}(x(t_1), x(t)) \\ \vdots \\ \text{Cov}(x(t_M), x(t)) \\ 1 \end{bmatrix} \quad (10)$$

In this relation, the Lagrange multiplier allocated to limitation is represented by λ . Subsequence to indicating w_i weights, the $x(t)$ approximation () and also the error variance can be computed by Eqs. (6) and (10).

The covariance calculation has most computational burden in Eq. (10). So, this work utilized an efficient estimation for covariance of Gaussian procedure, in order to abate the computation burden of this optimization. In this estimation, the covariance function is estimated by covariogram as follows [31]:

$$\text{Cov}(x(t_i), x(t_j)) = C(d_{ij}) \quad (11)$$

In which, d_{ij} is Euclidean space of t_i and t_j . This distance is decreased to $d_{ij} = |t_i - t_j|$ in our 1-D problem. Moreover, $C(\cdot)$ denotes covariogram that is covariance function based on Euclidean distance. Multitude functions are introduced in several works as covariance function for this goal, containing: the linear function, the exponential function, the Gaussian functions as well as the spherical function [32]. the von Karman covariance model [33] that is a high performance function is regarded in this paper for covariance computing. Mathematical definition of this model is as follow:

$$C(r) = \frac{\sigma^2}{2^{v-1}\Gamma(v)} (r/a)^v K_v(r/a) \quad (12)$$

In this equation, $\Gamma(\cdot)$ and K_v determine respectively the gamma and edited Bessel function for second kind of order $0 \leq v \leq 1$ [34]. Also, the lag and correlation length are indicated by r and a , respectively. Moreover, variance is shown with σ^2 . The $C(r)$ that is based on r/a is depicted in Fig. 6 for various v amounts. Value of 0.5 is accepted for v , in this work that causes to exponential auto covariance model. Besides, optimization for

achieving a amount is done by Shark Smell Optimization (SSO) algorithm represented in [35]. This optimization is aimed to minimize error of variance. In [35] the SSO method is thoroughly described. Suggested IKIM consists of KIM and von Karman covariance function expressed by Eq. (12), also shark smell optimization utilized for optimization of a amount.

Parsing process of $x(t)$ with suggested EMD equipped to IKIM is stated step-by-step in following:

- Step 1: Put $x_1(t) = x(t)$;
- Step 2: Obtain the max amount of $x_1(t)$ and higher envelope of $u_e(t)$ by suggested IKIM;
- Step 3: Obtain min value of $x_1(t)$ and lesser envelope of $l_e(t)$ by suggested IKIM;
- Step 4: Obtain the regional average in form of $m(t) = [u_e(t) + l_e(t)]/2$.
- Step 5: Set $x_1(t) = x_1(t) - m(t)$, then through checking IMF features indicate that $x_1(t)$ is IMF or not. Iterate 2–5 steps till turn into IMF and save achieved IMF.
- Step 6: Put $x(t) = x(t) - x_1(t)$;
- Step 7: Iterate 1–6 paces till achieving whole IMFs as well as $x(t)$ residues.

Fig. 7 depicts the outputs that achieved by exerting of suggested EMD based on IKIM for shown case study in Fig. 2. Also, these results are contrasted with outputs of customary EMD based on CS. Obviously; the authenticity of suggested EMD enveloping is more that customary EMD. Also, spurious over and undershoots of created envelopes by customary EMD are dramatically reduced in comparison to created envelopes by suggested EMD. The authenticity of IMF elements can be improved through producing more precise envelopes in suggested EMD.

4. Proposed detection method

If active methods are used to identify the island conditions of DGs in the microgrid system, the microgrid system becomes unstable in island conditions. So, for all DGs in microgrid, the active method is not a good option. In available passive methods, island conditions are identified by

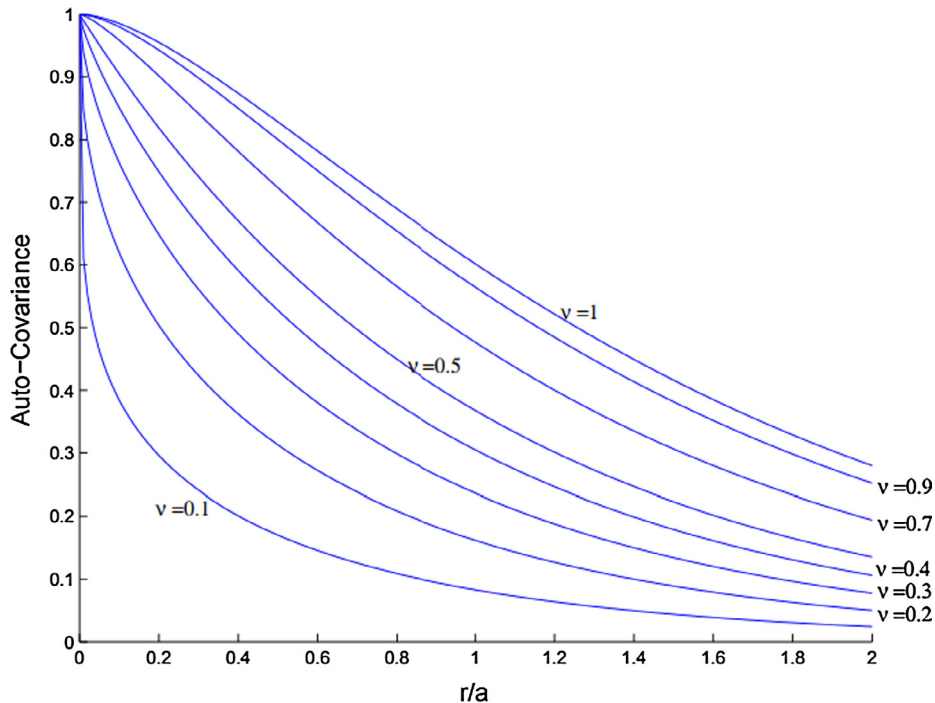


Fig. 6. $C(r)$ in terms of r/a for different values of v .

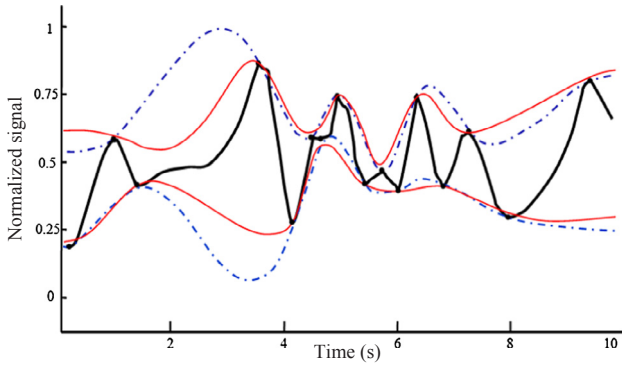


Fig. 7. Comparison of envelopes generated by conventional EMD using cubic spline (dash-dot blue line) with envelopes generated by the proposed EMD using Improved Kriging Interpolation Method (IKIM) (solid red line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

measuring one or variable, but this method has some disadvantageous, which is: Large non-detection zones, fail detection, threshold selection, etc. In this paper, with the aim of improving the mentioned deficiencies, it has helped to improve conditions by variation of state variables in two-dimensional spaces. In this paper, the two-dimensional space, by variation the paths for choosing state variables, it is clearly seen that the paths of variables are completely unlike from each other in both the islanding and non-islanding condition. The process is that state variables are expert to the learning device, and then the island conditions are founded.

4.1. Measured signal classification

Because of islanding event, transient conditions are created, that are identified by learning the parameters of state variables. In a two-dimensional space, by identifying the independent and sensitive parameters, the best signal is chosen for explore the proposed relay. Frequency voltage, power, and THD signal to find the island are selected from [15] and [20]. State parameters in this paper are IMF coefficient of KEMD of mentioned indices which are measured locally and saved all events. These variables are sorted by their type and relation. These parameters are formulated according to the equations as follows:

$$X_1 = [\Delta\delta/\Delta t \quad \Delta f/\Delta t] \quad (13)$$

$$X_2 = [\Delta P/\Delta t \quad \Delta Q/\Delta t] \quad (14)$$

$$X_3 = [\Delta V_d/\Delta t \quad \Delta V_q/\Delta t] \quad (15)$$

$$X_4 = [THD_v \quad THD_i] \quad (16)$$

In this section, for the saved signals, the mentioned above equation is used to select the IMF coefficients of KEMD.

4.2. Best signal selection

In order to identify the reliable islanding conditions through the KEMD theory, proper signal are chosen. Standard deviation values are exploited to choice the best signal for both islanding and non-islanding modes. The signal $x(n)$ can be decomposed in terms of IMF at $J = [0 \ 1 \ 2 \ \dots \ j-1]$ levels and the residual coefficient at $j = 0$ level. By a α_{coeff} matrix, mentioned coefficients are calculated as follows:

$$\alpha_{coeff} = [c_0 \ d_0 \ d_1 \ d_2 \ \dots \ d_{j-2} \ d_{j-1}] \quad (17)$$

The vector of IMF coefficients can be computed for all island states. IMF matrix are classified as $N \times J$ matrices in which N indicates number of islanding condition tests and J represents number of decomposition

level. An islanding signals standard deviation (ISSD) amount is obtained based on the above matrix for any coefficient. The mean square error for the i^{th} decomposition level coefficient is obtained as follows:

$$ISE_{dk} = \frac{1}{N} \sum_{i=1}^N \sum_{l=1}^M \frac{(E_{dk}(\Delta t_i, l) - \overline{E_{dk}}(\Delta t_i))^2}{M} \quad (18)$$

where, Δt_i is i^{th} discrete time, M is number of islanding test events, $E_{dk}(\Delta t_i, l)$ is IMF value of signals in i^{th} discrete time and l^{th} islanding events, $\overline{E_{dk}}(\Delta t_i)$ is the average value of E_{dk} for all islanding conditions, which is defined as:

$$\overline{E_{dk}}(\Delta t_i) = \frac{1}{M} \sum_{l=1}^M (E_{dk}(\Delta t_i, l)) \quad (19)$$

As well, the islanding signals standard deviation for i -th IMF level is computed by:

$$ISSD_{dk} = \sqrt{ISE_{dk}} \quad (20)$$

The difference between the value of IMFs of islanding and non-islanding conditions that represent across *mean square error (CIMSE)* value is calculated. The standard deviation of the mean energy amount in island state for all non-islanding state is called CINSSD that formulated as follows.

$$CINSSD_{dk} = \sqrt{\frac{1}{N} \sum_{i=1}^N \sum_{h=1}^L \frac{(E_{dk}(\Delta t_i, h) - \overline{E_{dk}}(\Delta t_i))^2}{L}} \quad (21)$$

where, L is the number of non-islanding test events. It should be noted that the amount of *ISSD* is bigger than *CINSSD* and this difference indicates that a method for classifying an unknown condition is should be proposed. An acceptable value for all conditions of the *CINSSD* to *ISSD* ratio is computed and arranged. In order to identify the algorithm, appropriate signal is chosen that has a great *CINSSD* to *ISSD* ratio.

As can be seen from Table 1, The IMF₂ value of active and reactive power variation signals has the maximum value of *CINSSD* /*ISSD*. In Fig. 8, the type of mentioned signal for CHP is shown in island conditions. In all conditions, the signals have the same manner.

4.3. Support vector machine

In this section, some of basic information introduces about SVM

Table 1
ISSD and CINSSD analysis of microgrid signals.

Signal Name	Coefficient	CHP unit value		
		ISSD	CINSSD	CINSSD /ISSD
$\frac{\partial P}{\partial t} + j \frac{\partial Q}{\partial t}$	RE	2.73	11.87	4.3480
	IMF ₁	1.85	27.11	14.654
	IMF ₂	0.922	58.6507	63.648
	IMF ₃	1.56	6.1	3.9103
	IMF ₄	3.48	7.54	2.1667
	IMF ₅	1.1	3.63	3.3000
$\Delta V_d + j \Delta V_q$	IMF ₆	0.46	1.45	3.1522
	RE	0.64	27.0248	42.226
	IMF ₁	4.26	11.589	2.7204
	IMF ₂	2.646	9.3069	3.5173
	IMF ₃	0.45	4.062	9.0267
	IMF ₄	0.122	0.5185	4.2500
$THD_v + j THD_i$	IMF ₅	0.34	0.4955	1.4574
	IMF ₆	0.85	0.9578	1.1268
	RE	4.125	11.1295	2.6981
	IMF ₁	3.5230	10.8564	3.0816
	IMF ₂	2.1760	7.9760	3.6654
	IMF ₃	1.4532	0.8565	0.5894
	IMF ₄	0.1307	0.26695	2.0425
	IMF ₅	0.0890	0.02010	0.2258
	IMF ₆	0.0641	0.0817	1.2746

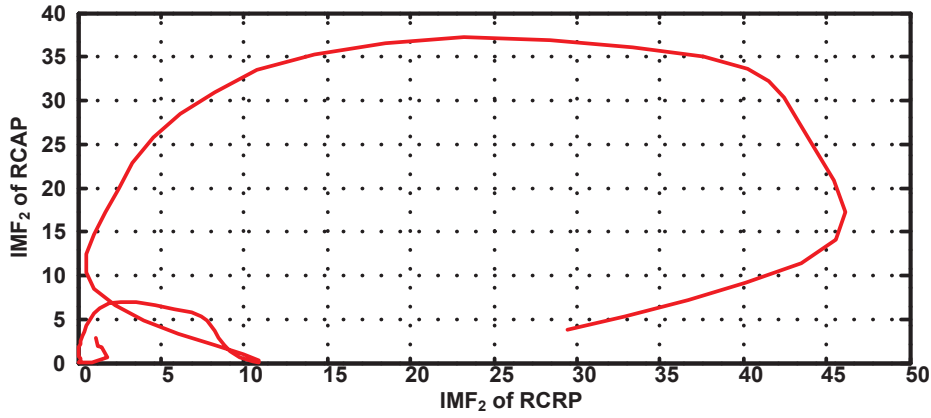


Fig. 8. Pattern of selected state variables in islanding conditions.

method. In SVM, $\{(x_1, y_1), \dots, (x_l, y_l)\}$ indicates training data sets, let $x_i \in R^n$ signed input space of the sample and has a same purpose value $y_i \in R^n$ for $i = 1, \dots, l$, where l is the size of the training data.

The support vector regression solves an optimization problem where, in this paper we consider the prediction error minimization as the following equation:

$$\frac{1}{2} \|w\|^2 + C \sum_{i=1}^l (\xi_i + \xi_i^*) \quad (22)$$

where

$$\begin{cases} y_i - \sum_{j=1}^n \sum_{i=1}^l w_j x_{ij} - b \leq \varepsilon + \xi_i \\ \sum_{j=1}^n \sum_{i=1}^l w_j x_{ij} + b - y_i \leq \varepsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases} \quad (23)$$

and

$$f(x) = \sum_{j=1}^n w_j x_j + b \text{ with } w \in R^n, b \in R \quad (24)$$

In this equation n represents the number of support vectors and b is the bias. w represents the weights for input (only the support vectors) to output. The trade-off between the complexity of the function f and the tolerance for error in the prediction of the function is shown by “ C ”. If the deviations of the machine’s prediction from the observed system behaviors that are smaller than ε , penalty factor is ignored. Also in above equation if errors in systems spares are less than ε we ignore it. In

Fig. 9 shown the ε = insensitive loss function.

Notice that it is not the traditional loss function (e.g., least square error), as the errors less than ε are ignored, in addition ξ_i and ξ_i^* definition the degree to which samples will be penalized with errors larger than ε shown the slack variables. When any error are smaller than ε in this case ξ_i and ξ_i^* are equal zero hence objective function eliminated. In this study between several kernel functions such as linear kernel functions, polynomial kernel functions radial basis functions (RBF) is used and defined as:

$$k(x_i, x) = \exp\{-\gamma |x - x_i|^2\} \quad (25)$$

The width of kernel signed with γ and varies between 0 and 1. In SVM model, two parameters (c and γ) are related together. One of the common way to estimate amount (c and γ) is separated the data in two section, namely a training data set and a validation data set. The accuracy of the model is given by prediction accuracy of this validation data. But it is a riskyly method.

One of the useful methods is Cross-validation for assessing how the results of statistical analysis can be generalized to an independent dataset. When the aim is prediction this method is suggested to use and one wants to estimate how accurately a predictive model will perform in practice [36]. The benefits of this method over repeated random sub sampling is that all the observations are used for both training and validation, and each observation is used for validation exactly once. Requires intense computation is disadvantage of this method. This study approves a k-fold cross-validation model that randomly barriers the original sample into k subsamples. An illustration of k-fold cross validation process is presented in Fig. 9. For testing the model a single

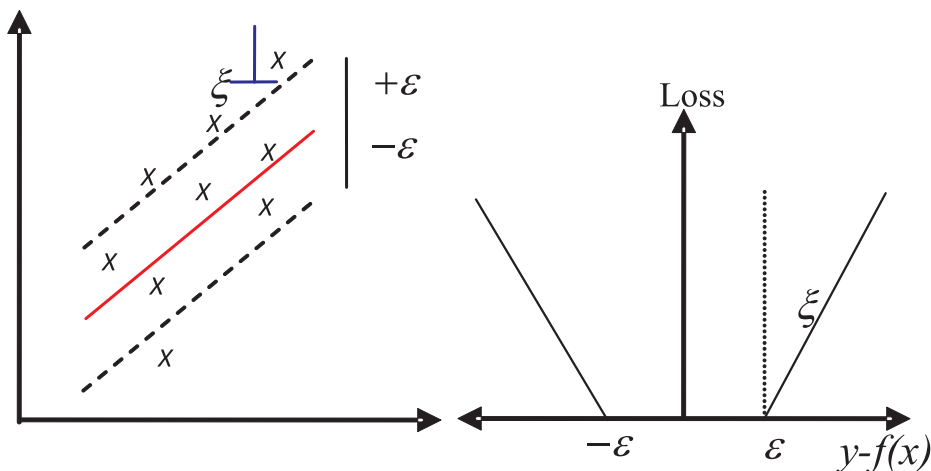


Fig. 9. Function of ε -insensitive loss and slack variable ξ .

subsample is protection as validation data, and the remaining $k-1$ subsamples are used as training data. In cross-validation technique time is repeated for k time, and each of them used once as the validation data. The k results from the folds can be averaged (or otherwise combined) to produce a single estimation.

4.4. Structure of proposed relay

Fig. 10 shows the structure of the proposed relay for CHP generator islanding detection. This technique uses rate of change of active and reactive power at the Power Common Coupling (PCC) point of CHP in study microgrid system. As can be seen in Fig. 10, in the next step, the variation of IMF₂ coefficient of KEMD for the inputs distributed generations is calculated. Using active and reactive signals, the value of IMF₂ variation of KEMD signals for these parameters are calculated (see Fig. 10). These values variation of KEMD coefficients are fed to the SVM for training and detecting the islanding conditions. If the islanding is detected, outputs of the proposed relay will be 1; otherwise, the islanding is not detected and the output will be 0. Using SVM system the proposed technique can overcome the problem of setting the detection thresholds inherent in the existing techniques.

The proposed relay training is one of the other important parts of the proposed relay. The number of input data and number of events data for training are two major factors that affects on the proposed relay performance.

- Number of input data (N_{ID}): The number of relay input data depends on the time interval for feeding of selected signal. As can be seen in Fig. 2, the transient time of the system is 0.2 s which is selected for the proposed method.
- Number of events for training (N_{EV}): another factor affecting the relay's performance is the number of island and non-island condition events for relay training. In the proposed method, number of events has changed between 10 and 90 percent of total data by steps of 5 percent.

In order to achieve the optimal condition, the number of input data and the number of event for training is selected by considering Receiver Operating Characteristic (ROC) and Area under the curve (AUC) [37]. One of the most appropriate criterions to appraising the efficiency of classifiers is the ROC. Some of utilities of this curve are specification of

proper operational condition of a certain classifier and indicating the most appropriate classifier through several of them. Number of true and false categorizations is the output of a classifier [37]. Here in order to determine the number of input data and number of events for relay training, the training data process is utilized. After signal selection process, 3-b, the IMF₂ coefficient energy of $\frac{\partial P}{\partial t} + j \frac{\partial Q}{\partial t}$ is selected, which have the high amount of ratio of intergroup signals to group signals, are assumed as input parameters. Then in order to choose the best number of input data and input events, training process is done step by step. In order to optimal training, the number of events has changed between 10 and 70 percent of all events by steps of 5 percent and the number of window time in each event has changed between 10 and 40 by steps of 2 percent. Proportional AUC value of ROC curves for each percent of input data and number of events is shown in Fig. 11. As can be seen the largest value of AUC is related to $N_{ID} = 32$ and $N_{EV} = 55$.

5. Simulation results

In this section, with the aim of demonstrating the benefits of the suggested method, simulation has been tested for both islands and non-island modes. In the following, the proposed identifying method is compared with other passive approach, and the advantages and precision of the proposed approach are shown.

5.1. Islanding conditions

In Fig. 12 for different loads in microgrid, IMF₂ variation of selected signals using KEMD transform for some islanding cases are depicted. Mentioned patterns are one of the most suitable applications for identifying of CHP generator. It should be noted that the state variables have the same behavior for all of the above islanding condition. In order to learning in the proposed relay, optimal percentage of the signals stored in the island state is used. Through sensitivity analysis of the suggested strategy choose some of the data. The results of the island identification method are compared with the results of other approaches in Fig. 14.

By analyzing the table results, it is seen that the correct detection rate with the suggested relay is high than the other methods. Therefore, the amount of error in detecting the island state using the suggested method is less and more reliable. The time to detect islands by using the proposed relay detection method is 0.15 sec, in other words, the

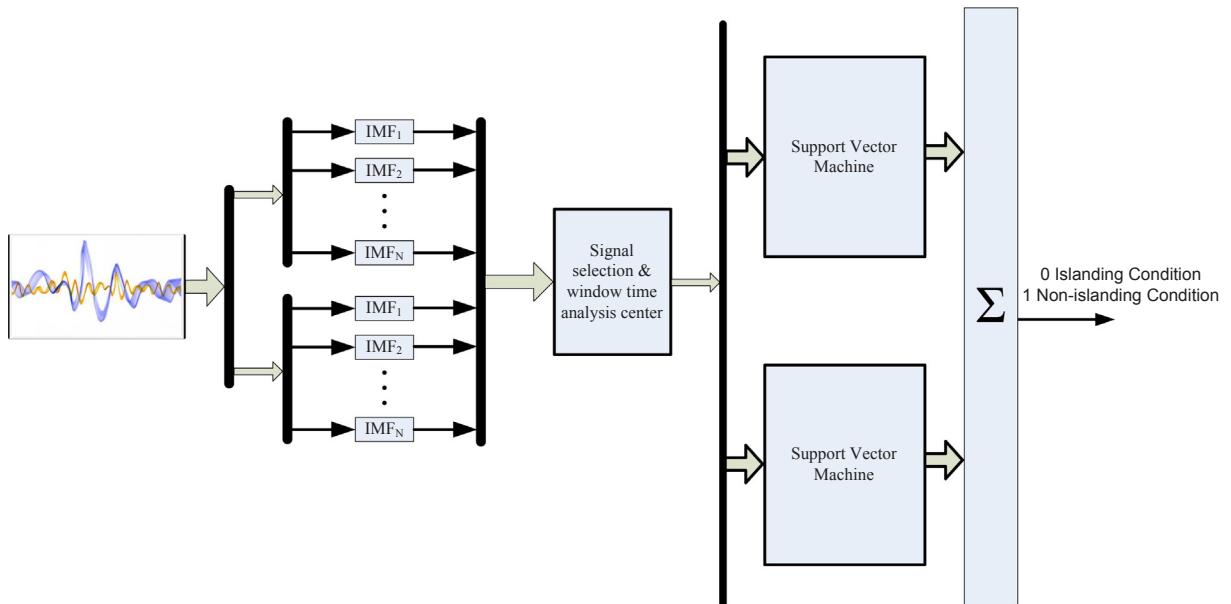


Fig. 10. Structure of the proposed relay for islanding detection.

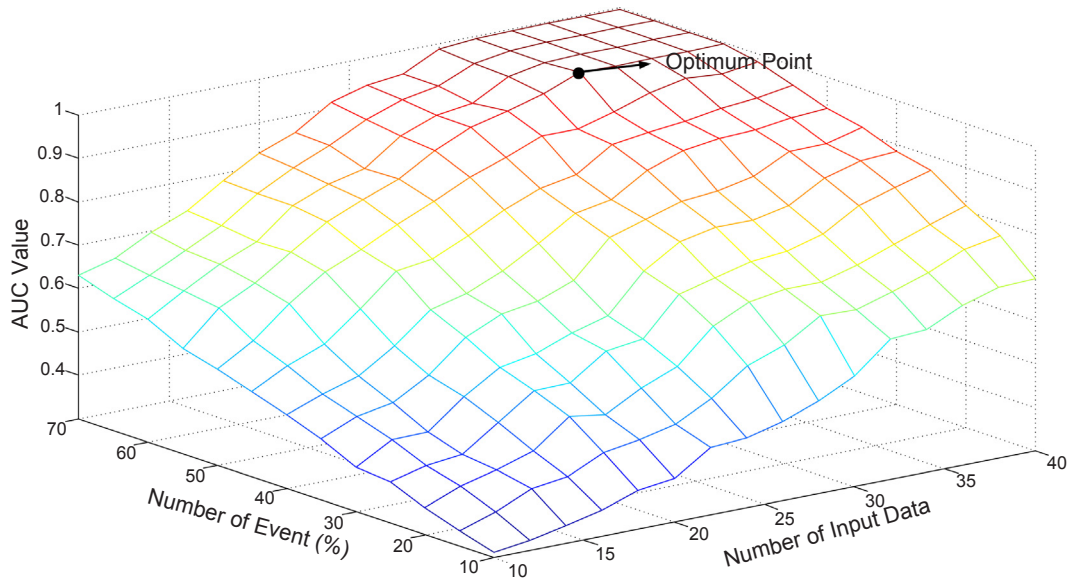


Fig. 11. AUC value of various number of input data and number of events.

detection time is much improved.

5.2. Non-islanding conditions

In this section, the results of the proposed approach during incidents, including the exchange of linear and nonlinear loads, large induction motor starting, and capacitor bank exchanging are compared with other methods of island identification. The prescribed non-island events for a variety of MG load modes are applied to the system with regard to injections or absorption active/reactive power. In all of these scenarios, the exchange happens at time 2 sec and all information is

stored for all of the conditions.

Results of the state variables paths for CHP generator is presented in Fig. 13. It should be noted that the outputs of the results in this figure are dependent on to linear and nonlinear switching, outage of generators, capacitor bank switching, and motor starting that shown in (a), (b), (c), and (d), respectively. Regarding the figure, it can be stated that the changes in the states are completely different in both island and non-island conditions. By analyzing the results, it is seen that the accuracy of the proposed method is higher than the other approach. The results of 1423 non-islanding cases using the proposed algorithm are presented in Fig. 14. Also, the results of the proposed relay in non-

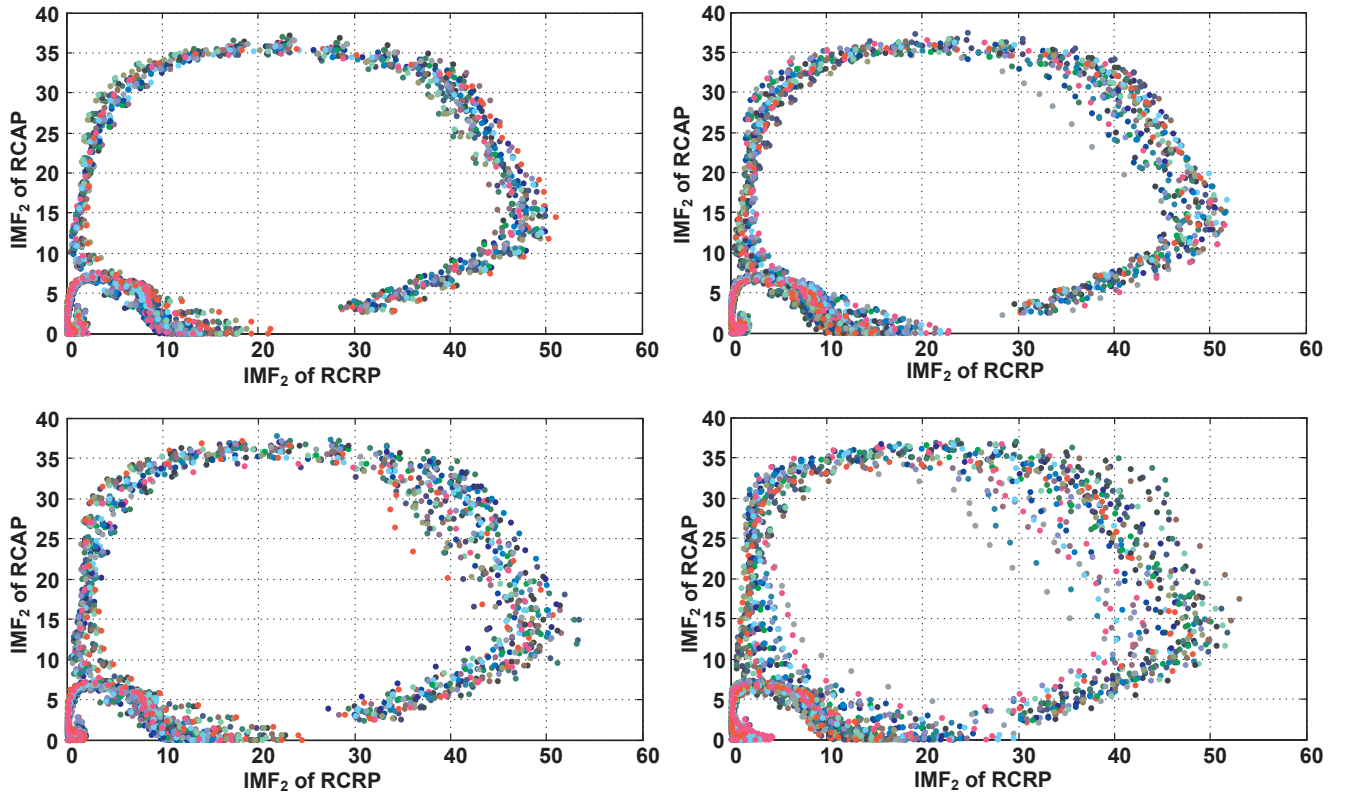


Fig. 12. Variations of the selected state variables for CHP in some islanding cases.

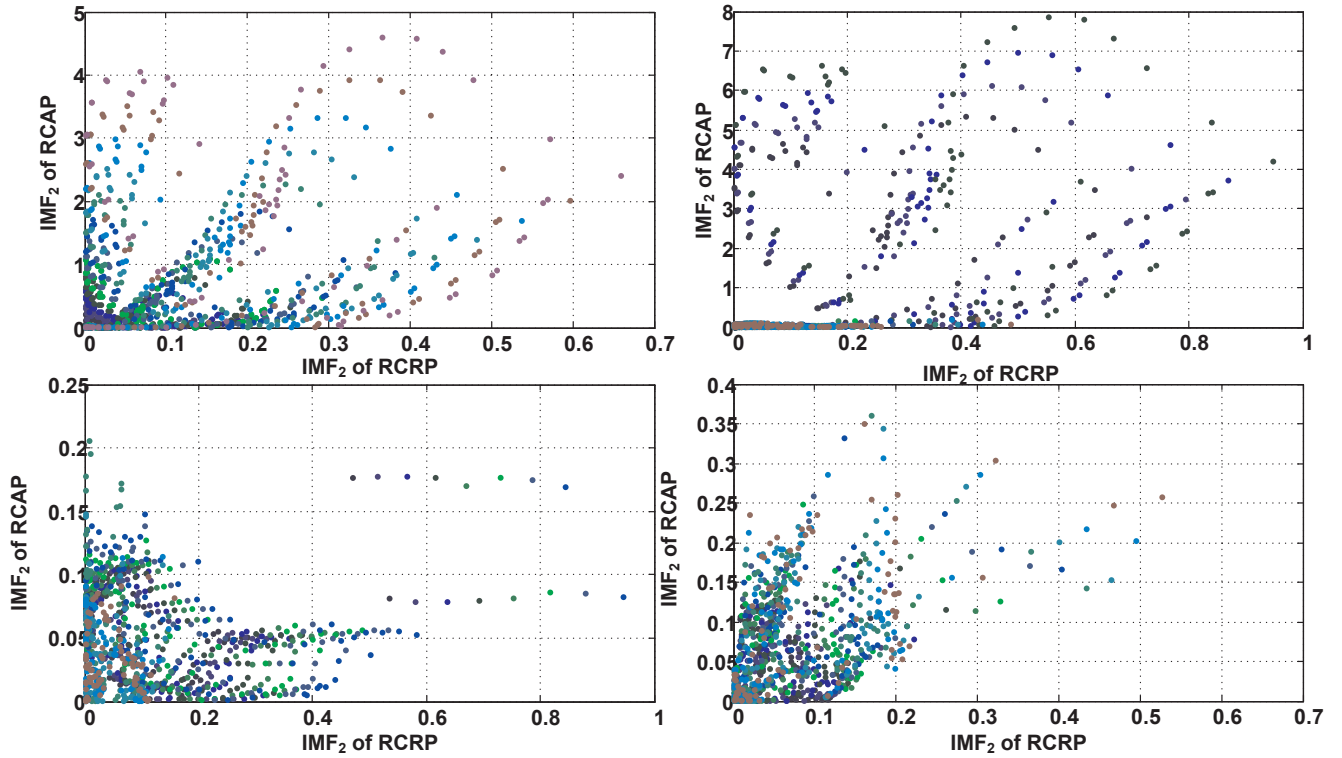


Fig. 13. Variations of the selected state variables in some non-islanding cases, (a) Linear and nonlinear load switching, (b) outage of generators, (c) capacitor bank switching, and (d) motor starting.

island conditions and the four active approaches mentioned in Fig. 14 are compared and no identification error has occurred with the suggested algorithm in non-island conditions. As a result, the proposed approach does not change in switching conditions. However, in four other identification approaches, there is a probability of identification error. The reliability of the identification method using the proposed strategy has considerably increased compared to other approach.

In order to show the performance of the proposed strategy in comparison to other methods, the compared results are shown in Fig. 15. In Fig. 14 (a), the number of islanding condition tests and the corrected detection of each relay is presented. In Fig. 14 (b) the percentage correct and false detection is plotted. The results of this figure is shows the performance of the proposed relay in comparison to other islanding detection. The correct detection of the proposed strategy is 99.34% and the false detection is zero. As can be seen, none detection zone of the proposed relay is very small than other methods.

5.3. Effects of noise on the proposed relay

The proposed relay is not trained with noisy voltage and current signals in the accessible results. Despite the noise in the signal, the proposed relay function is simulated to identify both island and non-island modes. The suggested islanding relay is tested by a set of noisy data with the aim of showing accurate performance and high identification accuracy of the state variable-based islanding relay. In noise condition, islanding relay performance is investigated and the SNR of voltage signal varies from 10 dB to 80 dB in the noisy condition of current (10 dB). The relay operation for CHP in noise conditions is depicted in Fig. 15. With precision in figure, it can be stated that the classification accuracy of the proposed relay decreases for the SNR of voltage signals less than 45 dB in the noisy condition of the current signal.

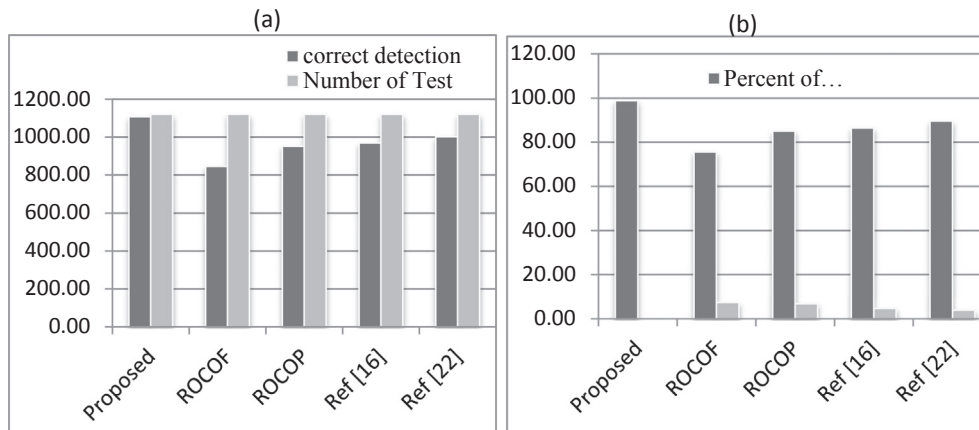


Fig. 14. The proposed relay comparison with the other active methods.

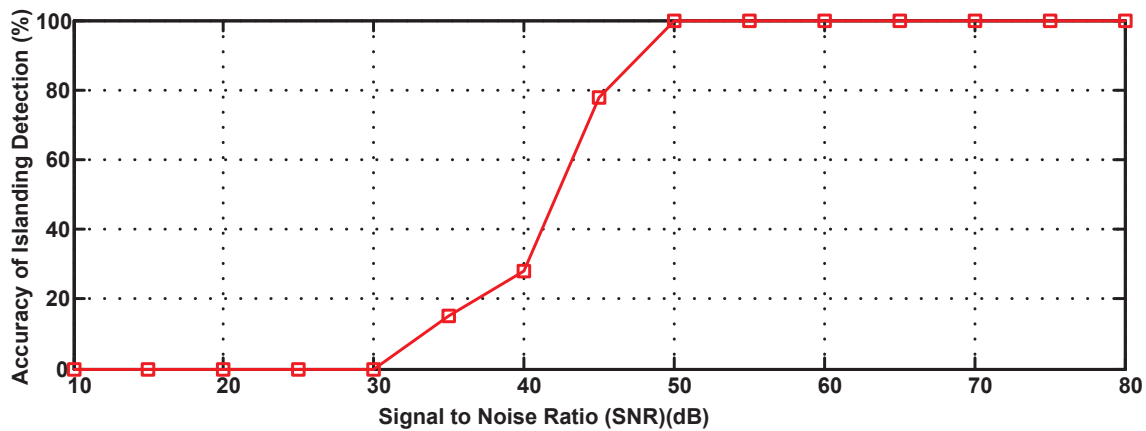


Fig. 15. Operation of the relay in the noisy situation.

5.4. Future work

Due to the expansion of distributed generation in power systems, the microgrids and smart grids will form the power system structure in the near future. In the other hand, a microgrid should be work in both islanding and autonomous situation with different control strategy. In order to change the control strategy in autonomous and islanding condition, it is necessary that the islanding condition is detected. In future works of islanding the islanding condition of each DG and whole of microgrid should be separated. Also, by developing of multi agent models, islanding detection should be detected by multi agent systems.

6. Conclusion

In this paper, an optimal passive technique was proposed to diagnose the islanding situation of CHP system in microgrid. This method employed KEMD analysis and state variables variation in transient conditions. By two-dimensional model and KEMD signals analysis, the proposed technique achieved to negligible non-detection zone (NDZ) and zero mistakes in detection. Moreover, by exploiting a SVM learning model, the proposed strategy resolved the threshold selection challenge. The proposed technique was applied to the microgrid test system composed of VSC-based DGs, wind turbine, CHP, and some linear and non-linear loads. The simulation results demonstrated the performance of the proposed relay in the islanding and non-islanding conditions. Compared with other passive methods, the proposed relay could detect almost all of the islanding conditions. Additionally, the fail detections of the proposed relay were zero. Small non-detection zone, zero fail detection, usability for all types of DGs, and Fast islanding detection were some of the advantages of the proposed islanding detection relay.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.applthermaleng.2018.09.028>.

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